

Quiz 2

✓ ① Which are hyperplanes?

A. a plane in A^3

$$\dim = 3 - 1 = 2 \checkmark$$

B. a point in A^1

$$\dim 1 - 1 = 0 \checkmark$$

C. a line in A^3 .

$$\dim 3 - 1 = 2$$

✗

a line has $\dim = 1$

D. a plane in A^4

$$\dim 4 - 1 = 3$$

✗

a plane has $\dim = 2$

✓ ② line l containing $H(2,4), N(2,-5)$

$$\begin{cases} x = \cancel{x}_H + t \cdot 0 \\ y = 4_H + t \cdot (-9) \end{cases} \Rightarrow \begin{cases} x = 2 + t \cdot 0 = 2 \\ y = 4 - 9t \end{cases}$$

$$\text{dir. vector: } \overrightarrow{HN} = N - H = (0, -9)$$

$$\text{Cartesian eq. } l: x - 2 = 0.$$

✓ ③ plane π containing $A(1,5,7)$, axis Ox

$$\begin{cases} x = 0 + 1 \cdot s + 1 \cdot t = s + t \\ y = 0 + 0 \cdot s + 5 \cdot t = 5t \\ z = 0 + 0 \cdot s + 7 \cdot t = 7t \end{cases}$$

$$(1,0,0)$$

$$\vec{OA} = (1,5,7)$$

$$O(0,0,0)$$

①

Ox axis $\Rightarrow$ origin $O(0,0,0)$

$\Rightarrow$ direction vector $i = (1,0,0)$.

$\Rightarrow OA = A - O = (1,5,7)$ another vector.
direction

Cartesian equation:

$\bullet (a,b,c) \perp (1,0,0)$ and $(1,5,7)$.
normal vector

$$n = (0, -7, 5)$$

$$(0, -7, 5) \cdot (1, 5, 7) = -35 + 35 = 0.$$

$$(0, -7, 5) \cdot (1, 0, 0) = 0.$$

$$\bar{n}: 0 \cdot (x - x_0) - 7(y - y_0) + 5(z - z_0) = 0$$

✓ (4) $a = ?$, $d = ?$ s.t. $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-3}{-2}$ is

contained in the plane $ax + y - 2z + d = 0$.
(-2)

dir. vector of the line: $(3, 2, -2)$

point: $P(2, -1, 3)$.

$$l \in \bar{n} : \Leftrightarrow \begin{cases} n \cdot m = 0 \\ P \in H \end{cases} \Leftrightarrow \begin{cases} a \cdot 3 + 1 \cdot 2 - 2 \cdot (-2) = 0 \\ -2 \cdot 2 - 1 - 2 \cdot 3 + d = 0 \end{cases}$$

$\Rightarrow -4 - 7 + d = 0$
 $\Rightarrow \boxed{d = +11}$

$\Rightarrow \boxed{a = -2}$

✓ (5) Relative pos. of lines:

$$l_1: \begin{cases} x = -3t \\ y = 2 + 3t \\ z = 1 \end{cases}$$

$$l_2: \begin{cases} x = 1 + 5s \\ y = 1 + 13s \\ z = 1 + 10s \end{cases}$$

dir. vector: $(-3, 3, 0)$ dir. vector: $(5, 13, 10)$

$\Rightarrow$ not proportional $\Rightarrow l_1 \nparallel l_2$.

(2)

$$z = 1 = 1 + 10s \Rightarrow 10s = 0 \Rightarrow s = 0.$$

$$x = 1 + 5 \cdot s = 1 + 5 \cdot 0 = 1. = -3t \Rightarrow t = -\frac{1}{3}.$$

$$l_1: y = 1 + 13 \cdot s = 1 + 13 \cdot 0 = 1. = 2 + 3 \cdot -\frac{1}{3} \quad \checkmark$$

$$l_2: y = 2 + 3 \cdot t = 2 - 3 \cdot \frac{1}{3} = 1 \quad \checkmark$$

$$\Rightarrow x = -3 \cdot -\frac{1}{3} = 1$$

$$\Rightarrow y = 1$$

$$\Rightarrow z = 1$$

$\Rightarrow$ both lines l_1 and l_2 produce sol. $(1, 1, 1)$

$\Rightarrow l_1 \cap l_2 \Rightarrow \boxed{D}$ incident in one point.

✓ (6). $A(1, 0, -1), B(0, 2, 3), C(-2, 1, 1), D(4, 2, 3)$.

dir. vectors
 $\vec{AB} = B - A = (-1, 2, 4)$
 $\vec{AC} = (-3, 1, 2)$
 } not proportional $\Rightarrow$
 not $\parallel \Rightarrow$ not collinear.
 $\boxed{D}$

coplanar if $V_{\text{parallelepiped}} = 0$.

$$\vec{AD} = (3, 2, 4)$$

$$\begin{vmatrix} -1 & 2 & 4 \\ -3 & 1 & 2 \\ 3 & 2 & 4 \end{vmatrix} \begin{matrix} \rightarrow \vec{AB} \\ \rightarrow \vec{AC} \\ \rightarrow \vec{AD} \end{matrix} = \dots = 36 - 36 = 0 \quad \boxed{A}$$

Setup 1: $\mathbb{R}^3$, ABCD tetrahedron.

$$K(B, \vec{AB}, \vec{AC}, \vec{AD})$$

✓ ⑦ eq. plane containing $\triangle BCD$ w.r. to K :

$$\vec{BP} = x\vec{AB} + y\vec{AC} + z\vec{AD}$$

$$[B]_K = (0, 0, 0).$$

$$\vec{BC} = \vec{BA} + \vec{AC} = -\vec{AB} + \vec{AC}$$

$$\Rightarrow [C]_K = (-1, 1, 0).$$

$$\vec{BD} = \vec{BA} + \vec{AD} = -\vec{AB} + \vec{AD}$$

$$\Rightarrow [D]_K = (-1, 0, 1).$$

$$n = \vec{BC} \times \vec{BD} = \begin{vmatrix} i & j & k \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = i \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} -$$

$$-j \begin{vmatrix} -1 & 0 \\ -1 & 1 \end{vmatrix} + k \begin{vmatrix} -1 & 1 \\ -1 & 0 \end{vmatrix} = i - j \cdot (-1) + k.$$

$$= i + j + k \Rightarrow (1, 1, 1)$$

$$\Rightarrow (BCD): 1(x-0) + 1(y-0) + 1(z-0) = 0$$

$$(BCD): x + y + z = 0.$$

✓ ⑧ Cartesian eq. of line AD w.r. to K :

$$\vec{AD} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ dir. vector.}$$

$$A(-1, 0, 0).$$

$$\vec{BA} = -1 \cdot \vec{AB} \Rightarrow (-1, 0, 0) = [A]_K.$$

$$\begin{cases} x = -1 + 0 \cdot t = -1 \\ y = 0 + 0 \cdot t = 0 \\ z = 0 + 1 \cdot t = t \end{cases}$$

what does not depend on t

$$\Rightarrow AD: \begin{cases} x + 1 = 0 \\ y = 0 \end{cases}$$

9. $K = (0, 3). \mathbb{H}^2.$

$K' = (0', 3').$

$[0]_K = \begin{bmatrix} -2 \\ -2 \end{bmatrix}, [i']_B = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, [j']_B = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$

$l: 3x + y - 2 = 0. (K)$

$l': \quad \quad \quad (K')?$

$M_{BB'} = \begin{bmatrix} 3 & 0 \\ -1 & -1 \end{bmatrix}$

$\begin{bmatrix} 3 & 0 & : & 1 & 0 \\ -1 & -1 & : & 0 & 1 \end{bmatrix} \xrightarrow{\frac{1}{3}L_1 + L_2 \rightarrow L_2} \begin{bmatrix} 3 & 0 & : & 1 & 0 \\ 0 & -1 & : & \frac{1}{3} & \phi \end{bmatrix} \xrightarrow{\begin{matrix} \frac{1}{3}L_1 \rightarrow L_1 \\ -L_2 \rightarrow L_2 \end{matrix}}$

$\sim \begin{bmatrix} 1 & 0 & : & \frac{1}{3} & 0 \\ 0 & 1 & : & -\frac{1}{3} & -\phi \end{bmatrix}$
 $M_{BB'}^{-1}$

$\Rightarrow M_{BB'}^{-1} = \begin{bmatrix} \frac{1}{3} & 0 \\ -\frac{1}{3} & 0 \end{bmatrix} \sqrt{3}$

$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 \\ -\frac{1}{3} & -\phi \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + [0]_K.$
 $= \begin{bmatrix} \frac{1}{3}x' \\ -\frac{1}{3}x' - y' \end{bmatrix} + \begin{bmatrix} -2 \\ -2 \end{bmatrix}$

$\Rightarrow \begin{cases} x = \frac{1}{3}x' - 2 \\ y = -\frac{1}{3}x' - 2 - y' \end{cases} \Rightarrow \begin{cases} x = 3x' - 2 \\ y = -x' - y' - 2 \end{cases}$

$l: 3x + y - 2 = 0 (K)$

$\downarrow$

$(K'): l': 3(3x' - 2) + (-x' - y' - 2) - 2 = 0$
 $l': 8x' - y' - 10 = 0$

Setup 2: $K(O, B), K'(O', B')$

✓ (10) CD
(param. eq.)

$$\begin{cases} x' = \\ y' = \\ z' = \end{cases}$$

$$[c]_{K'} = M_{B'B} ([c]_K - [O']_K)$$

$$M_{B'B} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{-L_2 \rightarrow L_2 \\ L_1 + L_3 \rightarrow L_1}} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{L_1 + L_3 \rightarrow L_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 3 & 1 & 0 & 1 \end{array} \right] \dots$$

$$\Rightarrow M_{B'B}^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & 1 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix}$$

$$[c]_{K'} = \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & 1 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix} \cdot \left(\begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right)$$

$$= \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & 1 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$[D]_{K'} = \dots = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

direction vector: $v' = [D]_{K'} - [C]_{K'} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$x' \rightarrow [C]_{K'}$ $C(0, -1, 0)$

$$\begin{cases} x' = 0 + t \cdot 1 \\ y' = -1 + t \cdot 1 \\ z' = 0 + t \cdot 1 \end{cases}$$

✓ (11) Cartesian eq: $(ABC) \leadsto K'$.

$$[A]_{K'} - [O']_{K'} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$$[A']_{K'} = M_{B'B} \cdot \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} = \dots = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$[B']_{K'} = M_{B'B} \cdot \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \dots = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$[C']_{K'} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

$\Rightarrow$ all points lie on the plane where $z' = 0$.

$z' = 0$ (cartesian eq. of the plane containing A, B, C)